

∴ The Pet in House 5 is Zebra

Final Answer: The Japanese has zebra as a pet and the Norwegian has mineral water as favorite drink.

43) Freedonia has 50 senators. Each senator ~~has~~^{is} either honest or corrupt. Suppose you know that at least one of the Freedonian senators is honest and that, given any two Freedonian senators, at least one is corrupt. Based on these facts, can you determine how many Freedonian senators are honest and how many are corrupt? If so, what is the answer?

[Soln:] Suppose, there are x honest Freedonian senators, where $x \geq 1$ but $x \leq 50$. Take a pair from these x senators. Both are honest. But this is a contradiction to the fact that given any two Freedonian senators, at least one is corrupt.

∴ Either $x=0$ or $x=1$. Since there is at least one honest Freedonian senator,

∴ $x=1$

Final Answer: 1 honest, 49 corrupt